WORKSHEET 6.7

The Fundamental Theorem, Part 2: Evaluating Integrals

Unit 6: Integration and Accumulation of Change

Key ideas

- **Fundamental Theorem, Part 2.** If $F' = f$, then $\displaystyle\int_a^b f(x)\,dx = F(b) - F(a)$.
- Write the antiderivative in brackets with the limits, then substitute: upper limit first.
- $\displaystyle\int_a^b f'(x)\,dx = f(b) - f(a)$: the integral of a rate of change is the net change. So $f(b) = f(a) + \displaystyle\int_a^b f'(x)\,dx$.
- For an absolute value or a piecewise function, split the interval where the formula changes.

Worked examples

Example 1 · A polynomial

Evaluate $\displaystyle\int_0^2 \left(3x^2 + 2x\right)\,dx$.

Solution. $\left[x^3 + x^2\right]_0^2 = (8 + 4) - 0$.

12.

Example 2 · A trigonometric function

Evaluate $\displaystyle\int_{\pi/6}^{\pi/2} \cos x\,dx$.

Solution. $\left[\sin x\right]_{\pi/6}^{\pi/2} = 1 - \dfrac{1}{2}$.

$\dfrac{1}{2}$.

Example 3 · A value from a rate

Given $f(1) = 4$ and $f'(x) = 2x + 3$, find $f(3)$.

Solution. $f(3) = 4 + \displaystyle\int_1^3 (2x + 3)\,dx = 4 + \left[x^2 + 3x\right]_1^3 = 4 + (18 - 4)$.

18.

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Practice

CORE

1. Evaluate $\int_1^3 2x\,dx$.

2. Evaluate $\int_1^4 \frac{1}{\sqrt{x}}\,dx$.

3. Evaluate $\int_0^{\ln 2} e^x\,dx$.

4. Evaluate $\int_1^2 \frac{3}{x^2}\,dx$.

EXAM LEVEL

5. Evaluate $\int_1^8 \sqrt[3]{x}\,dx$.

6. Evaluate $\int_0^{\pi/3} \sec x \tan x\,dx$.

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7. Let $f(x) = x^2$ for $x \leq 2$ and $f(x) = 4$ for $x > 2$. Evaluate $\displaystyle\int_0^5 f(x)\,dx$.

8. A particle has velocity $v(t) = 3t^2 - 6t$ and position $x(0) = 2$. Find $x(3)$.

Question 9 · Free-response practice

Answer each part. No calculator is permitted.

(a) Evaluate $\displaystyle\int_1^e \left(\frac{2}{x} + 1\right) dx$.

(b) A function f has $f(0) = 5$ and $f'(x) = 4\cos x$. Find $f\left(\frac{\pi}{2}\right)$.

(c) Find the area of the region under $y = 4 - x^2$ and above the x-axis for $0 \leq x \leq 2$.

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Worked solutions

1. $\Big[x^2\Big]_1^3 = 8.$

2. $\Big[2\sqrt{x}\Big]_1^4 = 4 - 2 = 2.$

3. $e^{\ln 2} - e^0 = 1.$

4. $\Big[-\frac{3}{x}\Big]_1^2 = -\frac{3}{2} + 3 = \frac{3}{2}.$

5. $\Big[\frac{3}{4}x^{4/3}\Big]_1^8 = \frac{3}{4}(16 - 1) = \frac{45}{4}.$

6. $\Big[\sec x\Big]_0^{\pi/3} = 2 - 1 = 1.$

7. $\displaystyle\int_0^2 x^2\,dx + \int_2^5 4\,dx = \frac{8}{3} + 12 = \frac{44}{3}.$

8. $2 + \Big[t^3 - 3t^2\Big]_0^3 = 2 + 0 = 2.$

9. (a) $\Big[2\ln x + x\Big]_1^e = (2 + e) - (0 + 1) = e + 1.$

   (b) $5 + \displaystyle\int_0^{\pi/2} 4\cos x\,dx = 5 + 4 = 9.$

   (c) $\Big[4x - \frac{x^3}{3}\Big]_0^2 = 8 - \frac{8}{3} = \frac{16}{3}.$

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